#In this code we will demonstrate the lemma 7.13. We will check that along the $\gamma 3$ curve the component nk is positive and along the $\gamma 4$ curve, nk changes sing exactly once.

#Define ñk

$$\tilde{n}k := \left((b^8 - b^6 + b^4)\, a^{10} - \frac{b^2 (b^8 - 2 b^6 + 8 b^4 - 2 b^2 + 1)\, a^8}{2} + \frac{b (b^4 - 4 b^2 + 1) (b^2 + 1)^3 a^7}{2} + \frac{(b^4 + 1) (b^8 + 8 b^4 + 1)\, a^6}{6} - \frac{b (b - 1)^2 (b + 1)^2 (b^2 + 1)^3 a^5}{2} - \frac{b^2 (b^8 + 2 b^6 - 2 b^4 + 2 b^2 + 1)\, a^4}{2} - \frac{b^3 (b^2 + 1)^3 a^3}{3} + a^2 b^6 - \frac{b^6}{3} \right):$$

#Define r3

$$r3 := \big(a^{22} b^{10} + (-6 b^{11} + 8 b^9) a^{21} + (15 b^{12} + 6 b^{10} + 30 b^8) a^{20} + (-18 b^{13} - 102 b^{11} + 162 b^9 + 74 b^7) a^{19} + (3 b^{14} + 198 b^{12} + 93 b^{10} + 600 b^8 + 145 b^6) a^{18} + (24 b^{15} - 96 b^{13} - 762 b^{11} + 1884 b^9 + 1380 b^7 + 252 b^5) a^{17} + (-294 b^{16} - 1740 b^{14} - 3162 b^{12} - 2920 b^{10} + 1908 b^8 + 1092 b^6 + 145 b^4) a^{16} + (24 b^{17} - 132 b^{15} - 2472 b^{13} - 8010 b^{11} + 3340 b^9 + 4692 b^7 + 1380 b^5 + 74 b^3) a^{15} + 3 b^2 (b^{16} - 580 b^{14} - 3862 b^{12} - 8784 b^{10} - 8307 b^8 - 1172 b^6 + 636 b^4 + 200 b^2 + 10) a^{14} + (-18 b^{19} - 96 b^{17} - 2472 b^{15} - 14520 b^{13} - 34416 b^{11} - 8268 b^9 + 3340 b^7 + 1884 b^5 + 162 b^3 + 8 b) a^{13} + (15 b^{20} + 198 b^{18} - 3162 b^{16} - 26352 b^{14} - 67416 b^{12} - 68232 b^{10} - 24921 b^8 - 2920 b^6 + 93 b^4 + 6 b^2 + 1) a^{12} - 6 b (b^{20} + 17 b^{18} + 127 b^{16} + 1335 b^{14} + 5736 b^{12} + 11692 b^{10} + 5736 b^8 + 1335 b^6 + 127 b^4 + 17 b^2 + 1) a^{11} + b^2 (b^{20} + 6 b^{18} + 93 b^{16} - 2920 b^{14} - 24921 b^{12} - 68232 b^{10} - 67416 b^8 - 26352 b^6 - 3162 b^4 + 198 b^2 + 15) a^{10} + (8 b^{21} + 162 b^{19} + 1884 b^{17} + 3340 b^{15} - 8268 b^{13} - 34416 b^{11} - 14520 b^9 - 2472 b^7 - 96 b^5 - 18 b^3) a^9 + (30 b^{20} + 600 b^{18} + 1908 b^{16} - 3516 b^{14} - 24921 b^{12} - 26352 b^{10} - 11586 b^8 - 1740 b^6 + 3 b^4) a^8 + (74 b^{19} + 1380 b^{17} + 4692 b^{15} + 3340 b^{13} - 8010 b^{11} - 2472 b^9 - 132 b^7 + 24 b^5) a^7 + (145 b^{18} + 1092 b^{16} + 1908 b^{14} - 2920 b^{12} - 3162 b^{10} - 1740 b^8 - 294 b^6) a^6 + (252 b^{17} + 1380 b^{15} + 1884 b^{13} - 762 b^{11} - 96 b^9 + 24 b^7) a^5 + (145 b^{16} + 600 b^{14} + 93 b^{12} + 198 b^{10} + 3 b^8) a^4 + (74 b^{15} + 162 b^{13} - 102 b^{11} - 18 b^9) a^3 + (30 b^{14} + 6 b^{12} + 15 b^{10}) a^2 + (8 b^{13} - 6 b^{11}) a + b^{12}\big):$$

#Apply the Möbius transformation on the rectangle [sqrt(3), 1+sqrt(2)) × [1, sqrt(3)) constructing r3

$$moebius := \left\{ a = \frac{\mathrm{sqrt}(3) + (1 + \mathrm{sqrt}(2))\, c}{1 + c},\; b = \frac{1 + \mathrm{sqrt}(3) \cdot d}{1 + d} \right\}:$$

r3 := collect(numer(expand(subs(moebius, r3))), c) :

#Compute the degree of r3 in c

```
degree(r3, c);
```

$$22 \quad \mathbf{(1)}$$

#Define the coefficients ci = coeff(r3, c, i)

```
for i from 0 to 22 do
  c[i] := coeff(`r3`, c, i);
od:
```

#Compute the all degree of ci in d

```
seq(degree(c[i], d), i = 0 ..22);
```

$$22, 22, 22, 22, 22, 22, 22, 22, 22, 22, 22, 22, 22, 22, 22, 22, 22, 22, 22, 22, 22, 22, 22 \quad \mathbf{(2)}$$

#From here we will analyze the signal variation of all coefficients ci

```
seq(seq(sign(evalf(coeff(c[j], d, i))), i = 22 ..0, -1), j = 22 ..0, -1);
```

$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1,$ **(3)**
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$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1$

#This demonstrate that the curve r3=0 does not intersect with ñk=0. As it cannot be contained on the left of ñk=0, implies that r3=0 are on the right. Therefore ñk is positive along the $\gamma 3$ curve.

#Define r4

$$r4 := \big(b^{12} a^{22} + (8 b^{13} - 6 b^{11}) a^{21} + (30 b^{14} + 6 b^{12} + 15 b^{10}) a^{20} + (74 b^{15} + 162 b^{13} - 102 b^{11} - 18 b^{9}) a^{19} + (145 b^{16} + 600 b^{14} + 93 b^{12} + 198 b^{10} + 3 b^{8}) a^{18} + (252 b^{17} + 1380 b^{15} + 1884 b^{13} - 762 b^{11} - 96 b^{9} + 24 b^{7}) a^{17} + (145 b^{18} + 1092 b^{16} + 1908 b^{14} - 2920 b^{12} - 3162 b^{10} - 1740 b^{8} - 294 b^{6}) a^{16} + (74 b^{19} + 1380 b^{17} + 4692 b^{15} + 3340 b^{13} - 8010 b^{11} - 2472 b^{9} - 132 b^{7} + 24 b^{5}) a^{15} + (30 b^{20} + 600 b^{18} + 1908 b^{16} - 3516 b^{14} - 24921 b^{12} - 26352 b^{10} - 11586 b^{8} - 1740 b^{6} + 3 b^{4}) a^{14} + (8 b^{21} + 162 b^{19} + 1884 b^{17} + 3340 b^{15} - 8268 b^{13} - 34416 b^{11} - 14520 b^{9} - 2472 b^{7} - 96 b^{5} - 18 b^{3}) a^{13} + b^{2} (b^{20} + 6 b^{18} + 93 b^{16} - 2920 b^{14} - 24921 b^{12} - 68232 b^{10} - 67416 b^{8} - 26352 b^{6} - 3162 b^{4} + 198 b^{2} + 15) a^{12} - 6 b (b^{20} + 17 b^{18} + 127 b^{16} + 1335 b^{14} + 5736 b^{12} + 11692 b^{10} + 5736 b^{8} + 1335 b^{6} + 127 b^{4} + 17 b^{2} + 1) a^{11} + (15 b^{20} + 198 b^{18} - 3162 b^{16} - 26352 b^{14} - 67416 b^{12} - 68232 b^{10} - 24921 b^{8} - 2920 b^{6} + 93 b^{4} + 6 b^{2} + 1) a^{10} + (-18 b^{19} - 96 b^{17} - 2472 b^{15} - 14520 b^{13} - 34416 b^{11} - 8268 b^{9} + 3340 b^{7} + 1884 b^{5} + 162 b^{3} + 8 b) a^{9} + 3 b^{2} (b^{16} - 580 b^{14} - 3862 b^{12} - 8784 b^{10} - 8307 b^{8} - 1172 b^{6} + 636 b^{4} + 200 b^{2} + 10) a^{8} + (24 b^{17} - 132 b^{15} - 2472 b^{13} - 8010 b^{11} + 3340 b^{9} + 4692 b^{7} + 1380 b^{5} + 74 b^{3}) a^{7} + (-294 b^{16} - 1740 b^{14} - 3162 b^{12} - 2920 b^{10} + 1908 b^{8} + 1092 b^{6} + 145 b^{4}) a^{6} + (24 b^{15} - 96 b^{13} - 762 b^{11} + 1884 b^{9} + 1380 b^{7} + 252 b^{5}) a^{5} + (3 b^{14} + 198 b^{12} + 93 b^{10} + 600 b^{8} + 145 b^{6}) a^{4} + (-18 b^{13} - 102 b^{11} + 162 b^{9} + 74 b^{7}) a^{3} + (15 b^{12} + 6 b^{10} + 30 b^{8}) a^{2} + (-6 b^{11} + 8 b^{9}) a + b^{10}\big) :$$

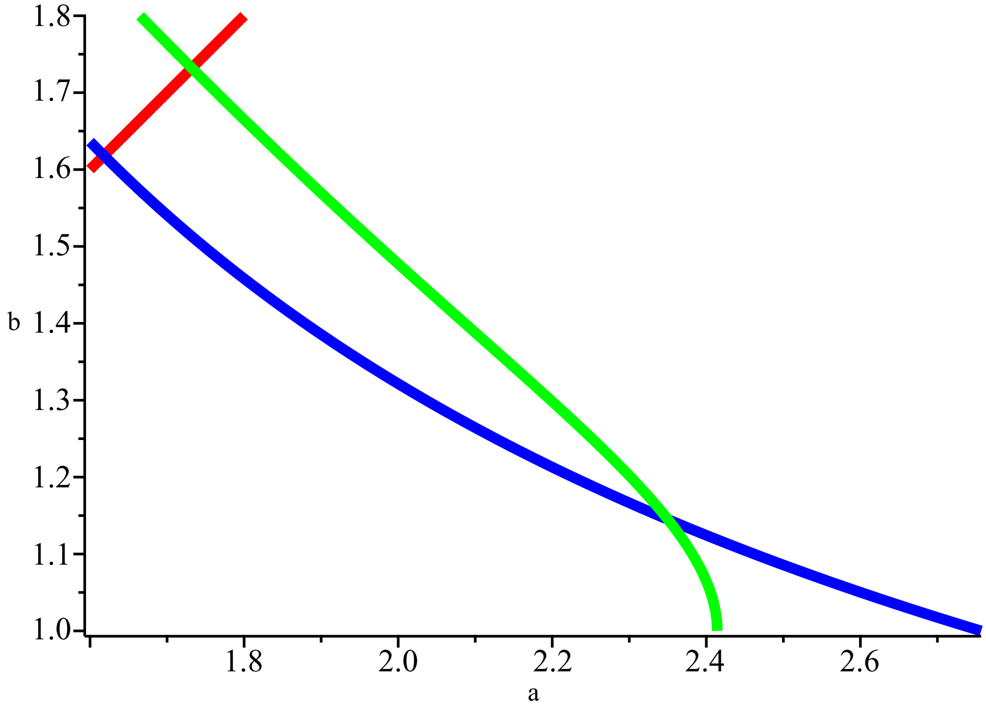
#Plot the curve ñk=0 and r4=0.

$r := a - b :$

```
with(plots) :

implicitplot( [r = 0, r4 = 0, ñk = 0], a = 1.6 .. 3, b = 1 .. (1.8),
        scaling = constrained,
        gridrefine = 2,
         thickness = 4,
        color = [red, blue, green],
        labels = ["a", "b"],
         title = "Graphic for ñk=0 and r4=0");
```

Graphic for ñk=0 and r4=0



$sturm\left(sturmseq(eval(r4, b=1), a)\ , a, \frac{5}{2}, 3\right)$; #number of roots in (2.5 , 3)

$$1 \qquad (4)$$

#*This check that*` $\gamma 4(1) > 2.5 > 1 + \mathrm{sqrt}(2)$. *How* $\frac{1+\mathrm{sqrt}(5)}{2} < \mathrm{sqrt}(3)$,

this demonstrate that the curves ñk = 0 and r4 = 0 intersect at lest one time. For uniqueness, we will calculate the resultant and apply the Sturm' algorithm.

$sturm\left(sturmseq(resultant(\text{ñk, r4, a}), b)\ , b, 1, \frac{1+\mathrm{sqrt}(5)}{2}\right)$; #number of roots in (1, v)

$$1 \qquad (5)$$

#The resultant of ñk and r4 with respect to a admit a uniqueness root in the interval where b are defined for r4=0.

$sturm(sturmseq(resultant(\text{ñk, r4, b}), a)\ , a, \mathrm{sqrt}(3), 1 + \mathrm{sqrt}(2))$;
#number of roots in (sqrt(3), 1+sqrt(2))

$$1 \qquad (6)$$

#The resultant of ñk and r4 with respect to b admit a uniqueness root in the interval where a are defined for ñk=0. This demonstrate the uniqueness.